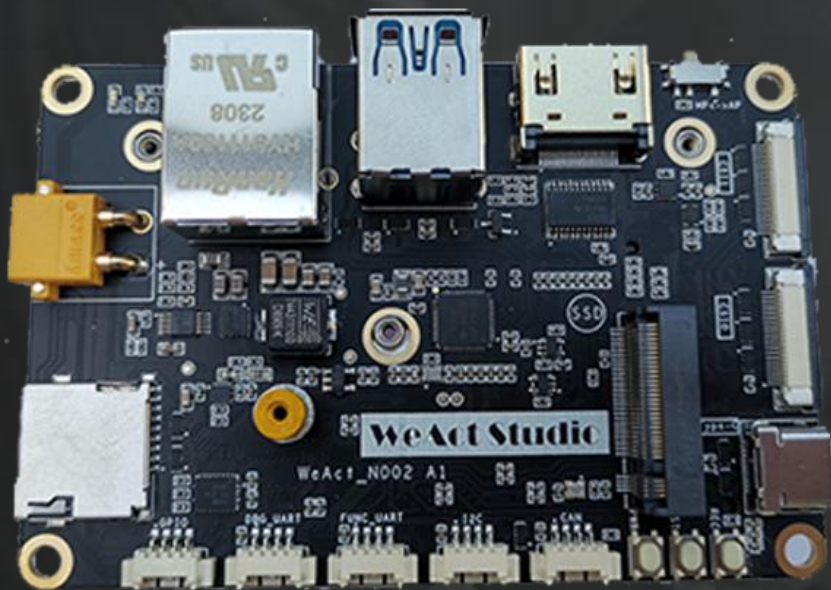




WeAct Studio

WEACT-N002

载板/底板

使用教程

WEAct Studio

目录

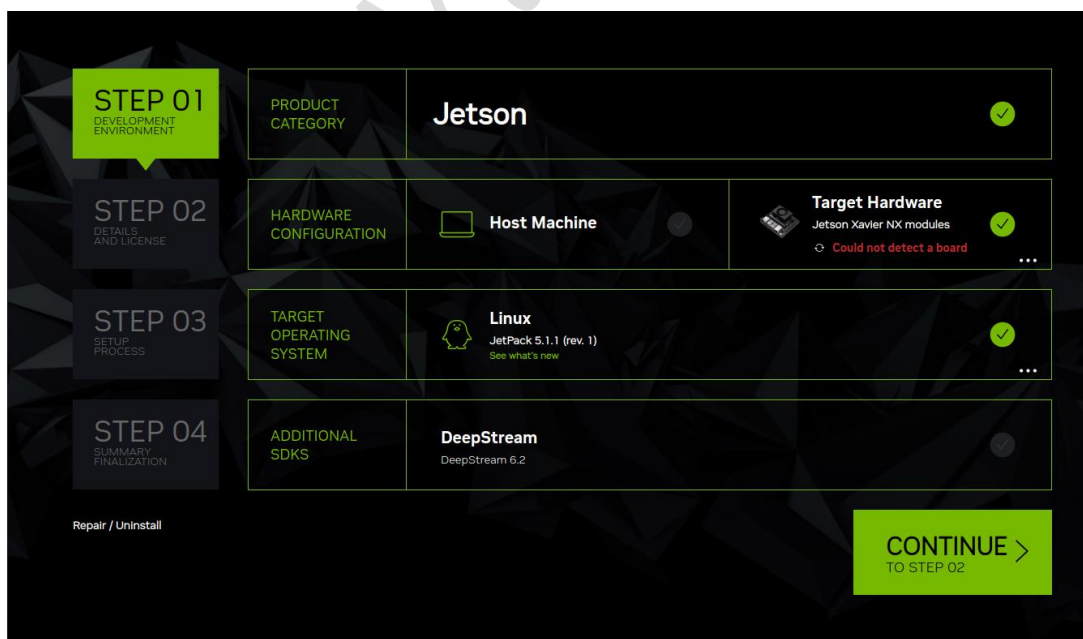
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REVISION HISTORY

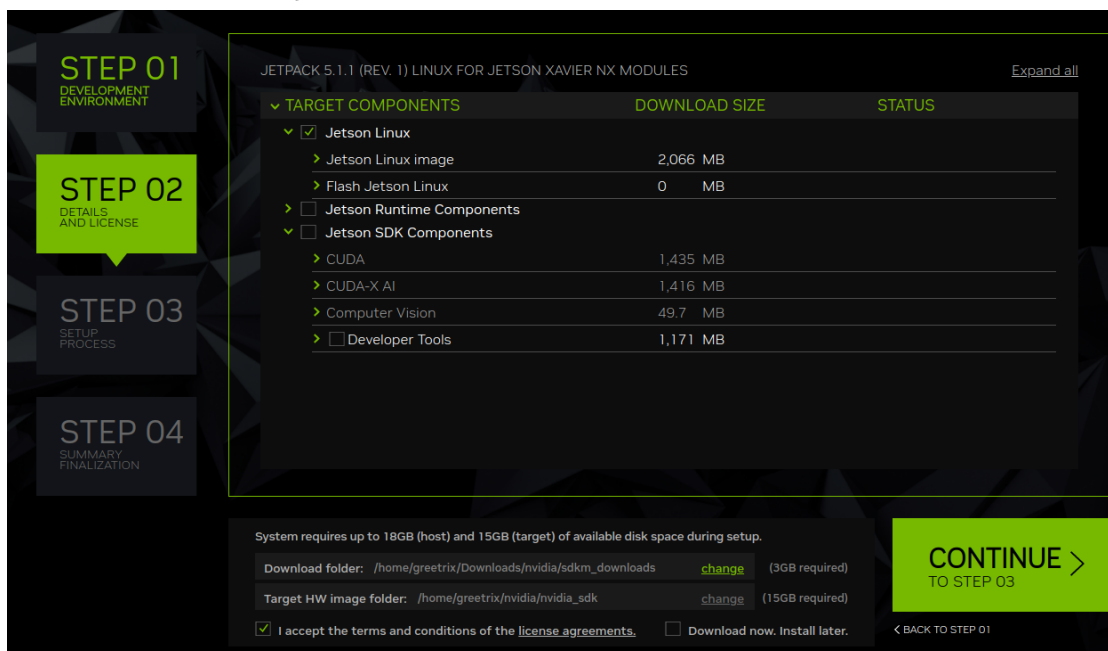
Draft Date	Revision	Description
2023.07.16	V1.0	1. 初始版本
2024.02.16	V2.0	1. 修订版本 2. 增加 GPIO 教程
2026.07.23	V3.0	1. 增加烧录脚本 2. Orin 支持 JP7

1. 搭建烧写环境

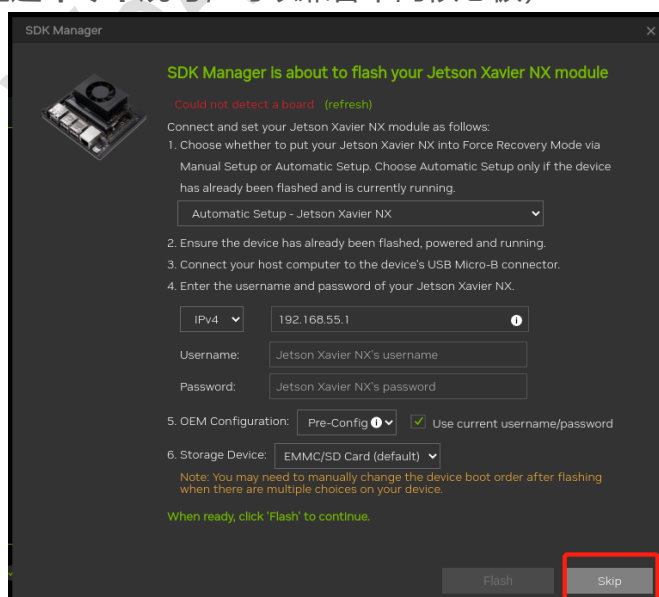
- a) 需要一台装有 **Ubuntu18.04** 以上的电脑作为 HOST 端给 Nano/NX 烧写，或者可以在 Windows 上安装 VMware 来安装 Ubuntu
- *Jetpack 6 需要 Ubuntu20.04 以上
 - *Jetpack 7 需要 Ubuntu24.04 以上
- VMware 上如何安装 Ubuntu18.04:
<https://blog.csdn.net/u012556114/article/details/82751089>
- b) 在 NVIDIA 下载最新的 **SDK-Manager** 并在 Ubuntu18.04 中安装（需要注册 NVIDIA 账号）
- SDK-Manager 下载地址: <https://developer.nvidia.com/nvidia-sdk-manager>
- c) 选择需要 **Target Hardware** 以及 **JetPack** 版本，**不勾选 HostMachine**（节省主机存储空间），DeepStream 根据组件需求勾选，这里以 **XavierNX** 为例选择，点击 **Continue**

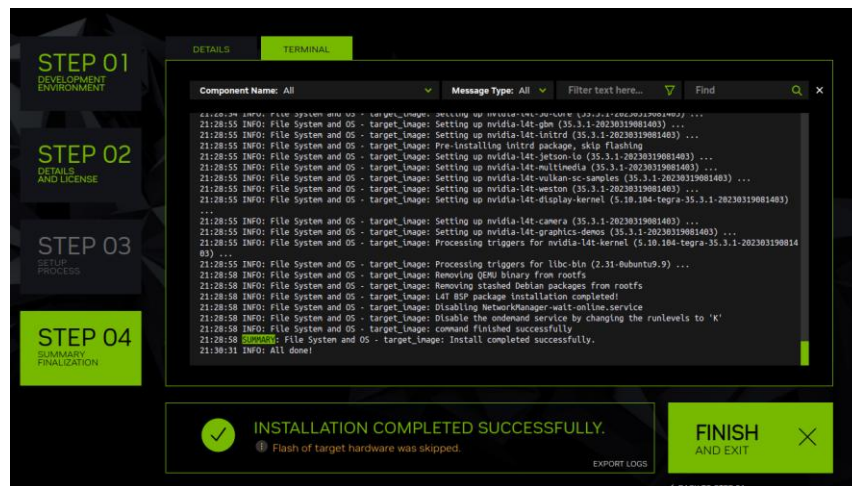


- d) 勾选 I accept the terms and conditions of the license agreements, 取消勾选 Jetson Runtime 及 SDK Components (后面安装组件会重新下载并一起安装), 点击 **CONTINUE** 进行下一步。

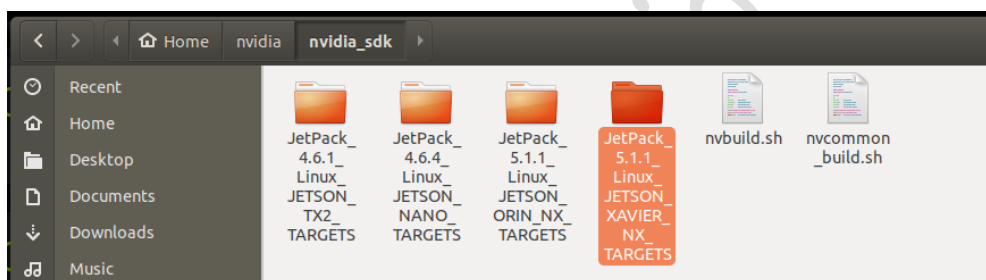


Note: 请在畅通的网络环境下进行下载以及安装, 下载或安装失败时, 可点击 **Retry** 继续, 直至全部状态为 **Installed** 并且显示绿色, 安装过程中会弹出联网烧写的信息, 选择 **Skip**。(后续步骤通过命令来烧写, 可以兼容不同核心板)





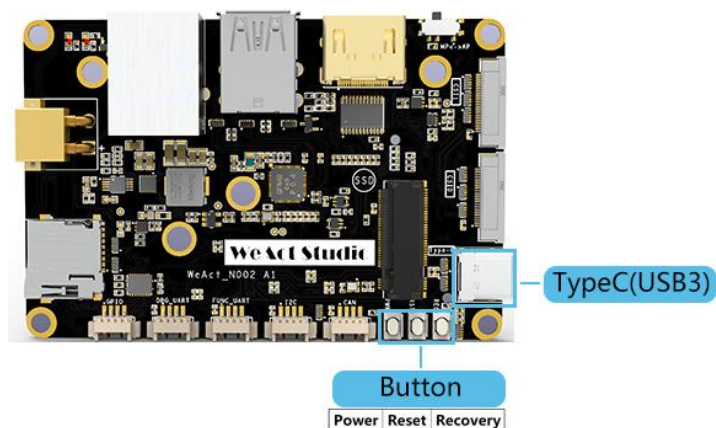
e) 安装成功后，会在 `~/nvidia/nvidia_sdk/` 下有相应版本烧写所需的文件



f) 在终端通过 `sudo apt-get install python` 安装 python 支持以便后续烧写环境。

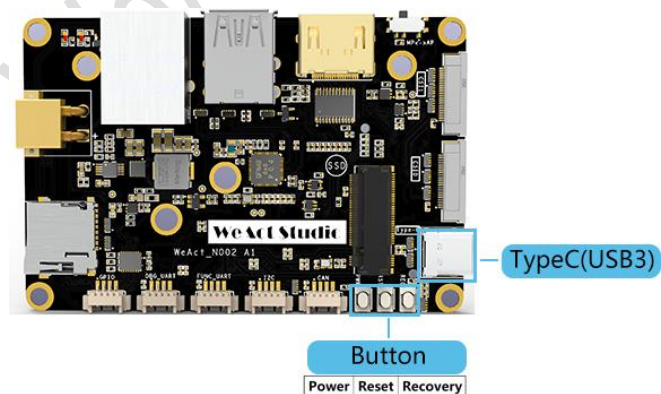
2. 核心板烧写系统

1. 使用 **USB Type-C** 线材连接载板上的 **USB Type-C** 接口



2. 将开机键拨至 **MP (手动开机)**，两种方式进入 Recovery 模式
 - a) 按住 **REC** 键，再按一下 **PWR** 键开机，松开 **REC** 键进入 Recovery 模式
 - b) 按一下 **PWR** 键开机，按住 **REC** 键，按一下 **RST** 键进入 Recovery 模式

此时 VMWare 右下角会出现 **NVIDIA** 的 **USB 驱动标志**，或者打开终端，输入 **lsusb** 命令，会发现 **Nvidia Corp**，同时风扇转速会到达最大。



```
_XAVIER_NX_TARGETS/Linux_for_Tegra$ lsusb
Bus 004 Device 001: ID 1d6b:0003 Linux Foundation 3.0 root hub
Bus 003 Device 004: ID 0e0f:0002 VMware, Inc. Virtual USB Hub
Bus 003 Device 003: ID 0e0f:0002 VMware, Inc. Virtual USB Hub
Bus 003 Device 006: ID 0955:7e19 NVidia Corp.
Bus 003 Device 002: ID 0e0f:0003 VMware, Inc. Virtual Mouse
Bus 003 Device 001: ID 1d6b:0002 Linux Foundation 2.0 root hub
Bus 001 Device 001: ID 1d6b:0002 Linux Foundation 2.0 root hub
Bus 002 Device 002: ID 0e0f:0002 VMware, Inc. Virtual USB Hub
Bus 002 Device 001: ID 1d6b:0001 Linux Foundation 1.1 root hub
```

3. 进入~/nvidia/nvidia_sdk/JetPack_5.1.1_Linux_JETSON_XAVIER_NX_TARGETS/Linux_for_Tegra (以 XavierNX 为例) , 打开 Terminal

- a) sudo apt-get install git
- b) git clone https://github.com/WeActStudio/WeAct_N002
 - i. sudo bash WeAct_N002/04_Script/weact_n002_flash.sh.

或

- c) git clone https://gitee.com/WeAct-TC/weact_n002
 - i. sudo bash WeAct_N002/04_Script/weact_n002_flash.sh

*Orin 烧录默认为 Super Mode, 如果版本不支持 Super Mode 或者不使用 Super Mode, sudo bash WeAct_N002/04_Script/weact_n002_flash.sh -n

*如果核心板为 Jetson Nano SD 版本或 Jetson XavierNX SD 版本, sudo bash WeAct_N002/04_Script/weact_n002_flash.sh -s

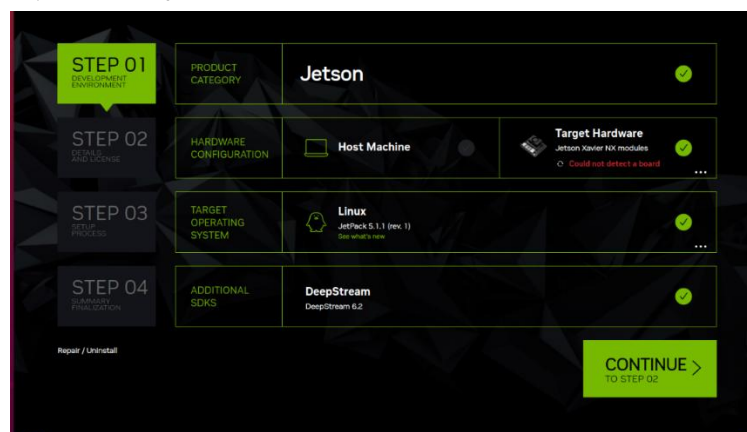
烧写系统成功后会有 Successfully!显示, 如下图所示。

```
File Edit View Search Terminal Help
t.encrypt [ 30416 bytes ]
[ 635.8133 ] [ ..... ] 100%
[ 636.0454 ] tegradevflash_v2 --write MEM_BCT mem_coldboot_sigheader.bct.encrypt
[ 636.0470 ] Bootloader version 01.00.0000
[ 636.0521 ] Writing partition MEM_BCT with mem_coldboot_sigheader.bct.encrypt [
198656 bytes ]
[ 636.0527 ] [ ..... ] 100%
[ 637.4597 ] tegradevflash_v2 --write MEM_BCT_b mem_coldboot_sigheader.bct.encrypt
pt
[ 637.4613 ] Bootloader version 01.00.0000
[ 637.4728 ] Writing partition MEM_BCT_b with mem_coldboot_sigheader.bct.encrypt
[ 198656 bytes ]
[ 637.4734 ] [ ..... ] 100%
[ 638.8778 ] Flashing completed

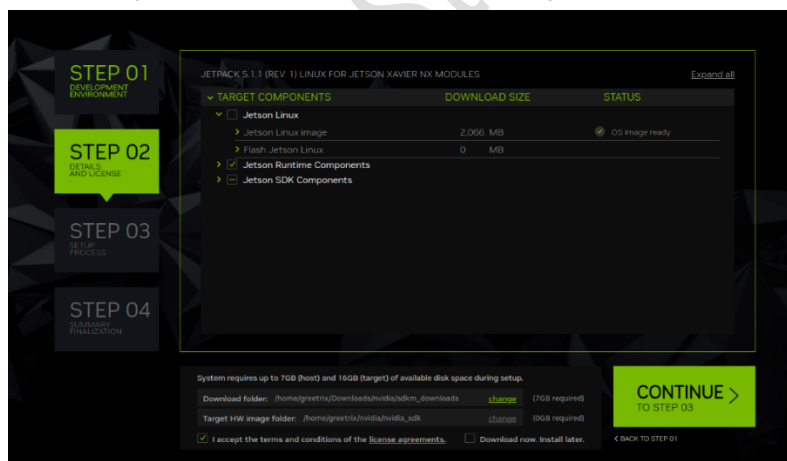
[ 638.8779 ] Coldbooting the device
[ 638.8825 ] tegrarcm_v2 --ismb2
[ 638.9047 ] tegradevflash_v2 --reboot coldboot
[ 638.9059 ] Bootloader version 01.00.0000
*** The target t186ref has been flashed successfully. ***
Reset the board to boot from internal eMMC.
```


3. 安装 NVIDIA 组件

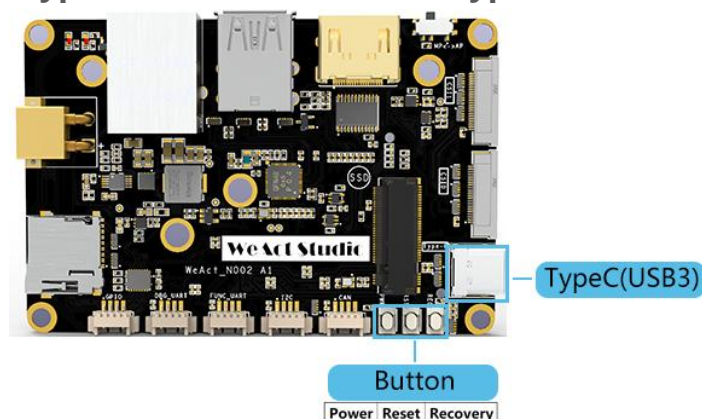
1. 选择需要 Target Hardware 以及 JetPack 版本，不勾选 HostMachine，这里以 XavierNX 为例选择，点击 Continue



2. 勾选所需要 SDK 组件，勾选 I accept the terms and conditions of the license agreements，点击 CONTINUE 进行下一步。

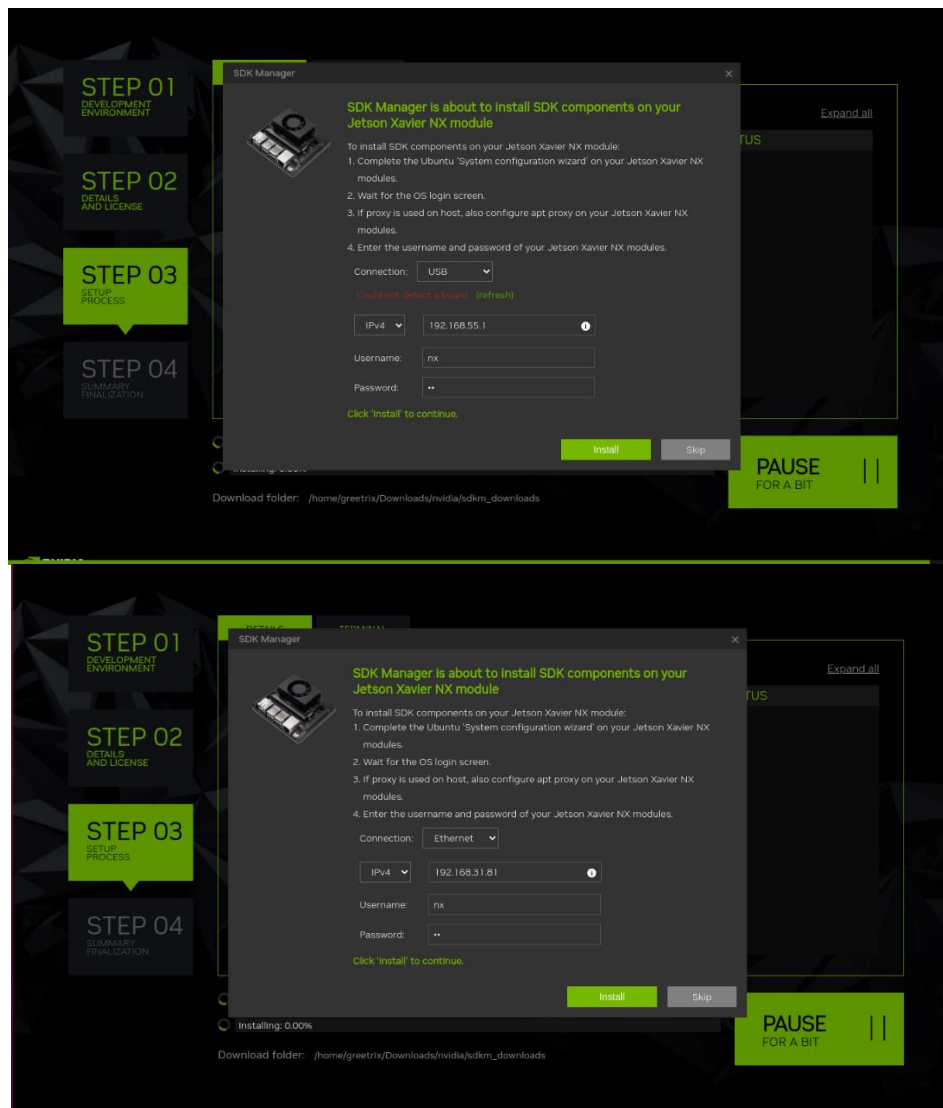


3. 使用 USB Type-C 线连接载板上的 USB Type-C 接口。



4. 将开机键拨至 MP（手动开机），按一下 PWR 键，正常开机并且进入系统，此时 VMWare 右下角会出现 NVIDIA 的 USB 驱动标志，或者打开终端，输入 `lsusb` 命令，会发现 **Nvidia Corp.**。

5. 输入 XavierNX 账号密码，XavierNX 端请保持联网状态，也可以用同局域网下的 IP（图 2，主机和 XavierNX 连接同一个路由器）



6. 等待安装完成即可。

4. 环境备份及镜像恢复

Note: 该备份仅用于 Jetson Nano/TX2NX/XavierNX

a) 系统安装在 EMMC, 以 TX2NX 为例

a) 参考第 2 章, 无论备份还是镜像烧写, 进入 Recovery 模式, 注意镜像较大, 请保证 Ubuntu 有充足的空间 (>40G) 。

b) **备份:** 这里以 TX2NX 为例 (其他设备参考上章内容修改 jetson 名称), 对核心板现有环境进行备份。进入 `~/nvidia/nvidia_sdk/JetPack_4.6_Linux_JETSON_TX2_TARGETS/Linux_for_Tegra`, 打开终端:

使用镜像备份命令: `sudo ./flash.sh -r -k APP -G backup.img jetson-xavier-nx-devkit-tx2-nx mmcblk0p1`, 等待备份完成即可, 此时目录下会有 backup.img 的镜像 (建议复制一份至其他位置备份), 此时**备份已经成功**。

```
greetrix@greetrix-virtual-machine:~/nvidia/nvidia_sdk/JetPack_4.6_Linux_JETSON_TX2_TARGETS/Linux_for_Tegra$ sudo ./flash.sh -r -k APP -G backup.img jetson-xavier-nx-devkit-tx2-nx mmcblk0p1
```

```
[ 9.1920 ] tegrarcm_v2 --boot recovery
[ 9.1966 ] Applet version 01.00.0000
[ 9.3692 ]
[ 10.3763 ] tegrarcm_v2 --isapplet
[ 10.3793 ] USB communication failed.Check if device is in recovery
[ 10.5068 ]
[ 10.8536 ] tegradevflash_v2 --iscpubl
[ 10.8565 ] Cannot Open USB
[ 11.3572 ]
[ 12.3617 ] tegrarcm_v2 --isapplet
[ 12.5109 ]
[ 12.5142 ] tegradevflash_v2 --iscpubl
[ 12.5163 ] Bootloader version 01.00.0000
[ 12.6843 ] Bootloader version 01.00.0000
[ 12.7463 ]
[ 12.7464 ] Reading partition
[ 12.7492 ] tegradevflash_v2 --read APP /home/greetrix/nvidia/nvidia_sdk/JetPack_4.6_Linux_JETSON_TX2_TARGETS/Linux_for_Tegra/backup.img
[ 12.7511 ] Bootloader version 01.00.0000
[ 12.9183 ] [.....] 100%
[ 2216.5426 ]
*** The [APP] has been read successfully. ***
Converting RAW image to Sparse image... greetrix@greetrix-virtual-machine
X2_TARGETS/Linux for Tegra$ k 4.6 Linux JETSON TX
```

c) **镜像烧写**：进入~/nvidia/nvidia_sdk/JetPack_4.6_Linux_JETSON_TX2_TARGETS/Linux_for_Tegra，将备份好的 backup.img 拷入 Linux_for_Tegra/bootloader/下，并重命名为 system.img，回到 Linux_for_Tegra 目录下，打开终端：

使用已有镜像烧写命令：**sudo ./flash.sh -r jetson-xavier-nx-devkit-tx2-nx mmcblk0p1**，等待烧写完成即可。

```
[ 18.0000 ] Writing partition spe-fw_b with spe_sigheader.bin.encrypt
[ 18.0298 ] [.....] 100%
[ 18.0790 ] Writing partition mb2 with nvtboot_sigheader.bin.encrypt
[ 18.1057 ] [.....] 100%
[ 18.1596 ] Writing partition mb2_b with nvtboot_sigheader.bin.encrypt
[ 18.1895 ] [.....] 100%
[ 18.2416 ] Writing partition mts-preboot with preboot_d15_prod_cr_sigheader.bi
n.encrypt
[ 18.2710 ] [.....] 100%
[ 18.6760 ] Writing partition mts-preboot_b with preboot_d15_prod_cr_sigheader.
bin.encrypt
[ 18.7053 ] [.....] 100%
[ 18.7467 ] Writing partition SMD with slot_metadata.bin
[ 18.7744 ] [.....] 100%
[ 18.9037 ] Writing partition SMD_b with slot_metadata.bin
[ 18.9302 ] [.....] 100%
[ 18.9658 ] Writing partition VER_b with emmc_bootblob_ver.txt
[ 18.9922 ] [.....] 100%
[ 19.0322 ] Writing partition VER with emmc_bootblob_ver.txt
[ 19.0592 ] [.....] 100%
[ 19.0966 ] Writing partition master_boot_record with mbr_1_3.bin
[ 19.1194 ] [.....] 100%
[ 19.1525 ] Writing partition APP with system.img
[ 19.1800 ] [.....] 016%
```

```
t.encrypt
[ 1888.6372 ] Bootloader version 01.00.0000
[ 1888.8013 ] Writing partition MB1_BCT with mb1_cold_boot_bct_MB1_sigheader.b
encrypt
[ 1888.8019 ] [.....] 100%
[ 1888.8706 ]
[ 1888.8837 ] tegradeflash_v2 --write MB1_BCT_b mb1_cold_boot_bct_MB1_sighead
bct.encrypt
[ 1888.8849 ] Bootloader version 01.00.0000
[ 1889.0452 ] Writing partition MB1_BCT_b with mb1_cold_boot_bct_MB1_sigheader
t.encrypt
[ 1889.0468 ] [.....] 100%
[ 1889.1180 ]
[ 1889.1181 ] Flashing completed

[ 1889.1181 ] Coldbooting the device
[ 1889.1436 ] tegradeflash_v2 --reboot coldboot
[ 1889.1449 ] Bootloader version 01.00.0000
[ 1889.3379 ]
*** The target t186ref has been flashed successfully. ***
Reset the board to boot from internal eMMC.
```

b) 系统安装在 SD 卡或者 SSD 的设备，直接通过主机拷贝 SD 卡或者 SSD 到其他存储介质即可

3. 备份和恢复系统 (ORIN NANO/NX)

Note: 该备份仅用于 Jetson Orin Nano/Orin NX

1. 使用和系统 Jetpack 版本一致的烧录环境，使用以下命令，等待备份完成即可
 - a) `sudo ./tools/backup_restore/l4t_backup_restore.sh -e nvme0n1 -b jetson-orin-nano-devkit-super (Super Mode)`
 - b) `sudo ./tools/backup_restore/l4t_backup_restore.sh -e nvme0n1 -b jetson-orin-nano-devkit (非 Super Mode)`

```
greetrix@ubuntu:~/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra$ sudo ./tools/backup_restore/l4t_backup_restore.sh -e nvme0n1 -b jetson-orin-nano-devkit-super
[sudo] password for greetrix:
Please install the Secureboot package to use initrd flash for fused board
# Entry added by NVIDIA initrd flash tool
/home/greetrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/tools/kernel_flash/tmp 127.0.0.1(rw,nohide,insecure,no_subtree_check,async,no_root_squash)
rpcbind: another rpcbind is already running. Aborting
Export list for localhost:
/home/greetrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/tools/kernel_flash/tmp 127.0.0.1
01:32:04.706 - Debug: Checking UFW status and NFS port rules...
01:32:04.758 - Debug: UFW is not active. NFS port is accessible.
01:32:04.758 - Debug: Checking VPN connections...
01:32:04.760 - Debug: IPsec service is not running.
01:32:04.761 - Debug: VPN routes none in tun|tap|ppp|vpn|gpd.
01:32:04.762 - Debug: No VPN connection detected.
/home/greetrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/tools/kernel_flash/l4t_initrd_flash_internal.sh --no-flash --initrd --showlogs --network usb0 jetson-orin-nano-devkit-super internal
*****
* Step 1: Generate rcm boot cmdline *
*****
ROOTFS_AB= ROOTFS_ENC= /home/greetrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/flash.sh --no-flash --rcm-boot jetson-orin-nano-devkit-super mmcblk0p1
#####

greetrix@ubuntu: ~/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_J...
nvbackup_partitions.sh: Start backing up nvme0n1p13 ...
1024+0 records in
1024+0 records out
67108864 bytes (67 MB, 64 MiB) copied, 1.87598 s, 35.8 MB/s
/mnt/images/temp : 0.05% (67108864 => 34356 bytes, /mnt/images/nvme0n1p13_bak.img)
nvbackup_partitions.sh: Success backing up nvme0n1p13 to nvme0n1p13_bak.img

nvbackup_partitions.sh: Start backing up nvme0n1p14 ...
6400+0 records in
6400+0 records out
419430400 bytes (419 MB, 400 MiB) copied, 11.8244 s, 35.5 MB/s
/mnt/images/temp : 0.04% (419430400 => 156111 bytes, /mnt/images/nvme0n1p14_bak.img)
nvbackup_partitions.sh: Success backing up nvme0n1p14 to nvme0n1p14_bak.img

nvbackup_partitions.sh: Start backing up nvme0n1p15 ...
7672+0 records in
7672+0 records out
502792192 bytes (503 MB, 480 MiB) copied, 14.01 s, 35.9 MB/s
/mnt/images/temp : 39.75% (502792192 => 199838179 bytes, /mnt/images/nvme0n1p15_bak.img)
nvbackup_partitions.sh: Success backing up nvme0n1p15 to nvme0n1p15_bak.img

nvbackup_partitions.sh: Backing up QSPI0...
64+0 records in
64+0 records out
67108864 bytes (67 MB, 64 MiB) copied, 3.35709 s, 20.0 MB/s
nvbackup_partitions.sh: Backup complete (after command prompt popping up)
Backup image is stored in /home/greetrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/tools/backup_restore/images
Operation finishes. You can manually reset the device
greetrix@ubuntu:~/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra$
```

2. 在备份完成的烧录环境，使用以下命令

- `sudo ./tools/backup_restore/l4t_backup_restore.sh -e nvme0n1 -r jetson-orin-nano-devkit-super (Super Mode)`
- `sudo ./tools/backup_restore/l4t_backup_restore.sh -e nvme0n1 -r jetson-orin-nano-devkit (非 Super Mode)`

```
greetrix@ubuntu:~/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/L
linux_for_Tegra$ sudo ./tools/backup_restore/l4t_backup_restore.sh -e nvme0n1 -r
jetson-orin-nano-devkit-super
[sudo] password for greetrix:
Please install the Secureboot package to use initrd flash for fused board
# Entry added by NVIDIA initrd flash tool
/home/greetrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Lin
x_for_Tegra/tools/kernel_flash/tmp 127.0.0.1(rw,nohide,insecure,no_subtree_check
,async,no_root_squash)
rpcbind: another rpcbind is already running. Aborting
Export list for localhost:
/home/greetrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Lin
x_for_Tegra/tools/kernel_flash/tmp 127.0.0.1
02:29:02.281 - Debug: Checking UFW status and NFS port rules...
02:29:02.334 - Debug: UFW is not active. NFS port is accessible.
02:29:02.335 - Debug: Checking VPN connections...
02:29:02.337 - Debug: IPsec service is not running.
02:29:02.338 - Debug: VPN routes none in tun|tap|ppp|vpn|gpd.
02:29:02.338 - Debug: No VPN connection detected.
/home/greetrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Lin
x_for_Tegra/tools/kernel_flash/l4t_initrd_flash_internal.sh --no-flash --initrd
--showlogs --network usb0 jetson-orin-nano-devkit-super internal
*****
*
* Step 1: Generate rcm boot cmdline *
*
*****
ROOTFS_AB= ROOTFS_ENC= /home/greetrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JET
SON_ORIN_NX_TARGETS/Linux_for_Tegra/flash.sh --no-flash --rcm-boot jetson-orin-n
ano-devkit-super mmcblk0p1
*****
# L4T BSP Information:
nvrestore_partitions.sh Restoring nvme0n1 with image nvme0n1p10_bak.img...
131072+0 records in
131072+0 records out
67108864 bytes (67 MB, 64 MiB) copied, 0.167374 s, 401 MB/s
nvrestore_partitions.sh Restoring nvme0n1 with image nvme0n1p11_bak.img...
163840+0 records in
163840+0 records out
83886080 bytes (84 MB, 80 MiB) copied, 0.210278 s, 399 MB/s
nvrestore_partitions.sh Restoring nvme0n1 with image nvme0n1p12_bak.img...
1024+0 records in
1024+0 records out
524288 bytes (524 kB, 512 KiB) copied, 0.00191661 s, 274 MB/s
nvrestore_partitions.sh Restoring nvme0n1 with image nvme0n1p13_bak.img...
131072+0 records in
131072+0 records out
67108864 bytes (67 MB, 64 MiB) copied, 0.184356 s, 364 MB/s
nvrestore_partitions.sh Restoring nvme0n1 with image nvme0n1p14_bak.img...
378186240 bytes (378 MB, 361 MiB) copied, 1 s, 378 MB/s
819200+0 records in
819200+0 records out
419430400 bytes (419 MB, 400 MiB) copied, 1.11363 s, 377 MB/s
nvrestore_partitions.sh Restoring nvme0n1 with image nvme0n1p15_bak.img...
496281088 bytes (496 MB, 473 MiB) copied, 5 s, 99.3 MB/s
982016+0 records in
982016+0 records out
502792192 bytes (503 MB, 480 MiB) copied, 5.60264 s, 89.7 MB/s
Erasing 65536 Kibyte @ 0 -- 100 % complete
67+1 records in
67+1 records out
67108864 bytes (67 MB, 64 MiB) copied, 68.2603 s, 983 kB/s
~
nvrestore_partitions.sh Successful restore of partitions on target board.
Operation finishes. You can manually reset the device
greetrix@ubuntu:~/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/L
linux_for_Tegra$
```


5. 系统迁移至 NVME 固态硬盘

Note: 该迁移仅用于 Jetson Nano/TX2NX/XavierNX

WeAct-N002 载板支持 PCIE3.0 X 4 插槽，支持 2242 NVME SSD M.2 接口固态硬盘，

a) NVME 固态硬盘配置：

- 1. 配置前确保系统能识别到 NVME 固态硬盘，终端命令：**sudo fdisk -lu**

```
Disk /dev/nvme0n1: 119.2 GiB, 128035676160 bytes, 250069680 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: gpt
```

- 2. 将 NVME 设置成 GPT 格式：

- i. 终端命令：**sudo parted /dev/nvme0n1** 进入 parted

```
tx2nx@tx2nx:~$ sudo parted /dev/nvme0n1
[sudo] password for tx2nx:
GNU Parted 3.2
Using /dev/nvme0n1
Welcome to GNU Parted! Type 'help' to view a list of commands.

(parted) █
```

- ii. 终端命令：**mklabel gpt** 将磁盘 label 设置为 gpt 格式

```
(parted) mklabel gpt
Warning: The existing disk label on /dev/nvme0n1 will be destroyed and all data on this disk will be lost. Do you want to continue?
Yes/No? Yes█
```

- iii. 终端命令：**mkpart logical 0 -1** 将磁盘 part 设置为 gpt 格式

```
(parted) mkpart logic 0 -1
Warning: The resulting partition is not properly aligned for best performance.
Ignore/Cancel? Ignore█
```

- iv. 终端命令：**print** 查看分区结果

```
(parted) print
Model: KBG40ZNS128G NVMe TOSHIBA 128GB (nvme)
Disk /dev/nvme0n1: 128GB
Sector size (logical/physical): 512B/512B
Partition Table: gpt
Disk Flags:

Number  Start   End     Size    File system  Name  Flags
  1      17.4kB  128GB   128GB                   logic
```

- v. 终端命令：**quit** 退出

- vi. 终端命令: **sudo fdisk /dev/nvme0n1**

```
(parted) quit
Information: You may need to update /etc/fstab.

tx2nx@tx2nx:~$ sudo fdisk /dev/nvme0n1

Welcome to fdisk (util-linux 2.31.1).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

Command (m for help):
```

- vii. Command (m for help): 输入 **N**, 选择增加新分区, 后面回车默认即可

```
Command (m for help): n
Partition number (2-128, default 2): 2
First sector (250067728-250069646, default 250068992):
Last sector, +sectors or +size{K,M,G,T,P} (250068992-250069646, default 250069646):

Created a new partition 2 of type 'Linux filesystem' and of size 327.5 KiB.
```

- viii. Command (m for help): 输入 **P**, 查看分区结果

Device	Start	End	Sectors	Size	Type
/dev/nvme0n1p1	34	250067727	250067694	119.2G	Linux filesystem
/dev/nvme0n1p2	250068992	250069646	655	327.5K	Linux filesystem

- ix. 终端命令: **quit** 退出

- x. 终端命令: **sudo mke2fs -t ext4 /dev/nvme0n1p1**, 格式化分区

```
tx2nx@tx2nx:~$ sudo mke2fs -t ext4 /dev/nvme0n1p1
mke2fs 1.44.1 (24-Mar-2018)
/dev/nvme0n1p1 contains a ext4 file system
      last mounted on / on Sun Dec 26 11:04:07 2021
Proceed anyway? (y,N)
```

- xi. 终端命令: **sudo mount /dev/nvme0n1p1 /mnt**, 成功 mount 则 NVME 配置成功

```
tx2nx@tx2nx:~$ sudo mount /dev/nvme0n1p1 /mnt
tx2nx@tx2nx:~$
```

b) NVIDIA Jetson 系统迁移 (!!!迁移前建议参考第 3 章进行系统备份):

- ✓ 下面以 TX2NX 为例, 其他设备替换命令中间的设备名称即可, 设备名称可参考上面命令
- 1. 终端命令: **git clone <https://github.com/jetsonhacks/rootOnNVMe>** 或者 **<https://github.com/jetsonhacks/rootOnNVMe>** 下载脚本
- 2. 进入 **rootOnNVMe** 文件夹, 终端命令: **./copy-rootfs-ssd.sh**, 复制系统文件

至 NVME SSD

```
tx2nx@tx2nx:/home/script/root0nNVMe-master$ ./copy-rootfs-ssd.sh
mount: /mnt: /dev/nvme0n1p1 already mounted on /mnt.
17,380,838 0% 2.40MB/s 0:00:06 (xfr#39, ir-chk=1015/44887)
```

➤ 3. 终端命令：./setup-service.sh 配置启动项

```
tx2nx@tx2nx:/home/script/root0nNVMe-master$ ./setup-service.sh
==== AUTHENTICATING FOR org.freedesktop.systemd1.reload-daemon ====
Authentication is required to reload the systemd state.
Authenticating as: tx2nx,, (tx2nx)
Password: Failed to reload daemon: Method call timed out
polkit-agent-helper-1: pam_authenticate failed: Authentication failure
Created symlink /etc/systemd/system/default.target.wants/setssdroot.service → /etc/systemd/system/setssdroot.service.
Service to set the rootfs to the SSD installed.
Make sure that you have copied the rootfs to SSD.
Reboot for changes to take effect.
```

- 4. 参考第 2 章，进入 Recovery 模式。
- 5. （烧录环境的 Ubuntu，参考前面章节）进入~/nvidia/nvidia_sdk/JetPack_4.6_Linux_JETSON_TX2_TARGETS/Linux_for_Tegra，打开终端：sudo ./flash.sh jetson-xavier-nx-devkit-tx2-nx nvme0n1p1 更新 EMMC 内部引导

```
greetrix@greetrix-virtual-machine:~/nvidia/nvidia_sdk/JetPack_4.6_Linux_JETSON_TX2_TARGETS/Linux_for_Tegra$ sudo ./flash.sh jetson-xavier-nx-devkit-tx2-nx nvme0n1p1
```

```
[ 37.3739 ] Coldbooting the device
[ 37.3775 ] tegradevflash_v2 --reboot coldboot
[ 37.3788 ] Bootloader version 01.00.0000
[ 37.5711 ]
*** The target t186ref has been flashed successfully. ***
Make the target filesystem available to the device and reset the board to boot from external nvme0n1p1.
```

- 6. 重启 TX2NX，终端命令：df -l，此时系统盘已经变为 NVME SSD，并且原有 EMMC 上系统已经成功迁移。

```
tx2nx@tx2nx:~$ df -l
Filesystem      1K-blocks    Used Available Use% Mounted on
/dev/nvme0n1p1 122547172 11949920 104329176 11% /
none            1578060      0    1578060  0% /dev
tmpfs           1962748      52    1962696  1% /dev/shm
tmpfs           1962748    20764    1941984  2% /run
tmpfs            5120         4        5116  1% /run/lock
tmpfs           1962748      0    1962748  0% /sys/fs/cgroup
tmpfs           392548      12    392536  1% /run/user/120
tmpfs           392548      0    392548  0% /run/user/1000
```

6. 系统迁移至 SD 卡

Note: 该迁移仅用于 Jetson Nano/TX2NX/XavierNX

a) SD 卡配置:

1. 配置前确保系统能识别到 SD 卡，终端命令：**sudo fdisk -lu**

```
Disk /dev/mmcblk1: 59.5 GiB, 63864569856 bytes, 124735488 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
```

2. 将 SD 卡设置成 GPT 格式:

i. 终端命令：**sudo fdisk /dev/mmcblk1**，进入 sd 卡配置

```
tx2nx@tx2nx:~/Desktop$ sudo fdisk /dev/mmcblk1
Welcome to fdisk (util-linux 2.31.1).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.
```

ii. 终端命令：**g**，新建 gpt 分区表

```
Command (m for help): g
Created a new GPT disklabel (GUID: E39DF30E-48FE-B041-A6FA-5EFAEC223CEA).
```

iii. 终端命令：**n**，新建分区，其他输入回车按默认值

```
Command (m for help): n
Partition number (1-128, default 1):
First sector (2048-124735454, default 2048):
Last sector, +sectors or +size{K,M,G,T,P} (2048-124735454, default 124735454):
```

iv. 终端命令：**w**，保存分区信息

```
Command (m for help): w
The partition table has been altered.
Calling ioctl() to re-read partition table.
Syncing disks.
```

v. 终端命令：**sudo mke2fs -t ext4 /dev/mmcblk1p1**，格式化分区

```

nx@nx-desktop:~$ sudo mke2fs -t ext4 /dev/mmcblk1p1
mke2fs 1.45.5 (07-Jan-2020)
Discarding device blocks: done
Creating filesystem with 15591675 4k blocks and 3899392 inodes
Filesystem UUID: fd266d8c-21ae-4dbd-89a6-6e5cd0f6be67
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376, 294912, 819200, 884736, 1605632, 26542
    08,
    4096000, 7962624, 11239424
Allocating group tables: done
Writing inode tables: done
Creating journal (65536 blocks): done
Writing superblocks and filesystem accounting information: done

```

vi. 终端命令: **sudo mount /dev/mmcblk1p1 /mnt**, 成功 mount 则 SD 卡配置成功

```

tx2nx@tx2nx:~/Desktop$ sudo mount /dev/mmcblk1p1 /mnt

```

b) NVIDIA Jetson 系统迁移 (!!!迁移前建议参考第 3 章进行系统备份):

- ✓ 下面以 TX2NX 为例, 其他设备替换命令中间的设备名称即可, 设备名称可参考上面命令
- 1. 终端命令: **git clone <https://github.com/jetsonhacks/rootOnNVMe>** 或者 **<https://github.com/jetsonhacks/rootOnNVMe>** 下载脚本
- 2. 修改 **copy-rootfs-ssd.sh** 文件, 注释掉 mount 命令

```

#!/bin/bash
# Mount the SSD as /mnt
# sudo mount /dev/nvme0n1p1 /mnt
# Copy over the rootfs from the SD card to the SSD
sudo rsync -axHAXX --numeric-ids --info=progress2 --exclude={"/dev/", "/proc/", "/sys/", "/tmp/",
"/run/", "/mnt/", "/media/", "/lost+found"} / /mnt
# We want to keep the SSD mounted for further operations
# So we do not unmount the SSD

```

- 3. 进入 **rootOnNVMe** 文件夹, 终端命令: **./copy-rootfs-ssd.sh**, 复制系统文件至 SD 卡

```

nx@nx-desktop:~/rootOnNVMe$ sudo bash ./copy-rootfs-ssd.sh
276,613,277 37% 50.12MB/s 0:00:05 (xfr#2089, ir-chk=1622/4960)

```

- 4. 参考第 2 章, 进入 **Recovery 模式**。
- 5. (烧录环境的 Ubuntu, 参考前面章节) 进入 **~/nvidia/nvidia_sdk/JetPack_4.6_Linux_JETSON_TX2_TARGETS/Linux_for_Tegra**, 打开终端:

将前面章节对应的刷机命令 **mmcblk0p1** 换成 **mmcblk1p1** 更新 EMMC 内部引导

- ◆ 如 **TX2NX-EMMC**: **sudo ./flash.sh jetson-xavier-nx-devkit-tx2-nx mmcblk1p1**

```

greetrix@greetrix-virtual-machine:~/nvidia/nvidia_sdk/JetPack_4.6_Linux_JETSON_TX2_TARGETS/Linux_for_Tegra$ sudo ./flash.sh jetson-xavier-nx-devkit-tx2-nx mmcblk1p1

```

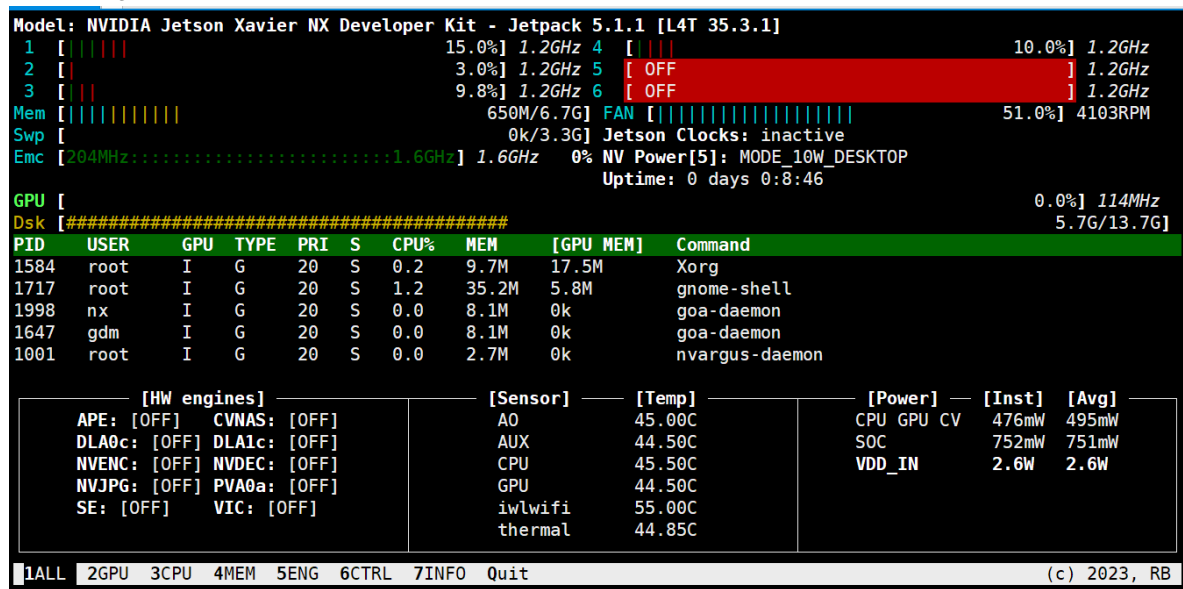
```
[ 30.4511 ] Coldbooting the device
[ 30.4521 ] tegradevflash_v2 --reboot coldboot
[ 30.4531 ] Bootloader version 01.00.0000
[ 30.6253 ]
*** The target t186ref has been flashed successfully. ***
Make the target filesystem available to the device and reset the board to boot from external mmcblk1p1.
```

- 6. 重启 TX2NX，终端命令：**df -l**，此时系统盘已经变为 SD 卡，并且原有 EMMC 上系统已经成功迁移。

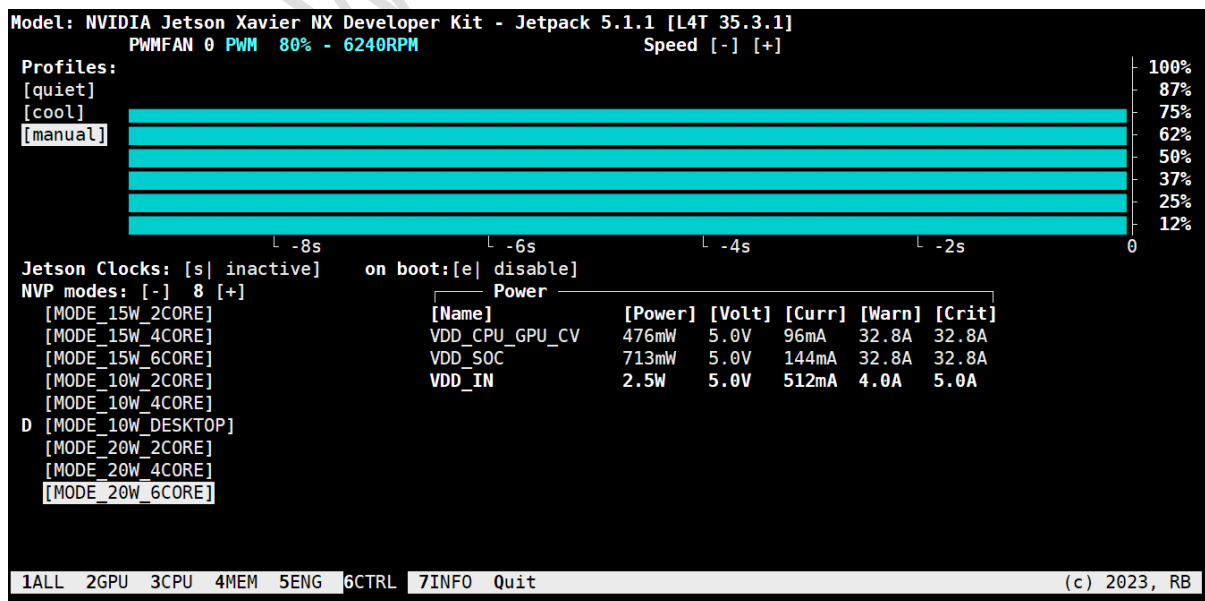
```
tx2nx@tx2nx:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
/dev/mmcblk1p1  59G   12G   44G   21% /
devtmpfs        1.6G     0   1.6G    0% /dev
tmpfs           1.9G   52K   1.9G    1% /dev/shm
tmpfs           1.9G   21M   1.9G    2% /run
tmpfs           5.0M   4.0K   5.0M    1% /run/lock
tmpfs           1.9G     0   1.9G    0% /sys/fs/cgroup
tmpfs           384M   12K   384M    1% /run/user/120
tmpfs           384M     0   384M    0% /run/user/1000
```

7. 安装 JTOP (控制风扇, 查看系统信息)

- 1. 打开终端, 通过下面命令安装 Jtop
 - a) sudo apt-get install python3-pip python3-dev -y
 - b) sudo -H pip3 install jetson-stats
 - c) sudo systemctl restart jtop.service
 - d) sudo jtop



- 2. 点击 6CTRL, 即可手动控制风扇转速及电源方案



8. 使用 CAN 进行通信

1. Tx2-NX/XavierNX/OrinNano/OrinNX 上集成了 1 个 CAN 控制器 CAN0，另外 WeAct Studio 的载板上设计了 1 个 CAN 收发器（CAN0），可直接挂载 CAN 物理总线使用。
2. Tx2-NX/XavierNOrinNX/OrinNanoX 自带 canbus 的驱动并集成到了镜像中，已经支持 canbus 无需多做处理。我们需要安装 canbus 模块。（在终端输入下面命令或者放入 rc.local 里面开启自启）

```
modprobe can      // 插入 can 总线子系统
modprobe can-raw   //插入 can 协议模块
modprobe can-bcm
modprobe can-gw
modprobe can_dev
modprobe mttcan    //真正的 can 口支持
```

3. 通过 **lsmod** 检查是否安装成功。

```
nvidia@localhost:~$ lsmod
Module                  Size  Used by
fuse                   103841  2
mttcan                  66251  0
can_dev                 13306  1 mttcan
can_gw                  10919  0
can_bcm                 16471  0
can_raw                 10388  0
can                     46600  3 can_raw,can_bcm,can_gw
zram                    26166  6
overlay                 48691  0
bcmhdhd                 934274  0
cfg80211                589351  1 bcmhdhd
spidev                  13282  0
nvgpu                  1575721  20
bluedroid_pm            13912  0
ip_tables               19441  0
x_tables                28951  1 ip_tables
```

- #### 4. 配置 canbus 属性，和串口的波特率设置类似。

```
sudo ip link set can0 type can bitrate 500000
sudo ip link set up can0
```

5. 通过 ifconfig 查看是否配置成功。

```
nvidia@localhost:~$ ifconfig
can0: flags=193<UP,RUNNING,NOARP> mtu 16
    unspec 00-00-00-00-00-00-00-00-00-00-00-00-00-00-00 txqueuelen 10  (UNSPEC)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
    device interrupt 131
```

6. 在一个终端通过 `cansend can0(can1) xxx` 命令来发送数据, 另一个终端通过 `candump can1(can0)` 完成实际信号收发测试

[illegible]

9. GPIO 使用

1. Jetson Nano/TX2NX/XavierNX/OrinNano/OrinNX 可直接通过 gpiod 控制 GPIO 输入输出

Jetpack 4

	Nano	TX2NX	Xavier NX
GPIO1	gpiochip0[149]	gpiochip0[105]	gpiochip0[133]
GPIO2	gpiochip0[216]	gpiochip0[76]	gpiochip0[148]
GPIO3	gpiochip0[194]	gpiochip0[18]	gpiochip1[20]
GPIO4	gpiochip0[38]	gpiochip0[13]	gpiochip0[105]

Jetpack 5

	XavierNX	Orin Nano/Orin NX
GPIO1	gpiochip1[105]	gpiochip0[105]
GPIO2	gpiochip1[118]	gpiochip0[144]
GPIO3	gpiochip2[16]	gpiochip0[85]
GPIO4	gpiochip1[84]	gpiochip0[43]

Jetpack 6/7

	Orin Nano/Orin NX
GPIO1	gpiochip0[105]
GPIO2	gpiochip0[144]
GPIO3	gpiochip0[85]
GPIO4	gpiochip0[43]

以 Orin NX Jetpack 7 GPIO1 为例, gpiochip0 是 Chip Group ID, 105 是 GPIO Number

2. 安装 gpiod/libgpiod-dev

- a) `sudo apt-get install gpiod`
- b) `sudo apt-get install libgpiod-dev`

3. GPIO 使用方法

- a) 设置高电平: `sudo gpiowrite gpiochip0 105=1`
- b) 设置低电平: `sudo gpiowrite gpiochip0 105=0`
- c) 读取: `sudo gpioread gpiochip0 105`

10. UART 使用

Jetpack 4

	Nano	TX2NX	Xavier NX
FUNC_UART	ttyTHS1	ttyTHS2	ttyTHS0

Jetpack 5

	Orin Nano/Orin NX	Xavier NX
FUNC_UART	ttyTHS0	ttyTHS0

Jetpack 6/7

	Orin Nano/Orin NX
FUNC_UART	ttyTHS1

1. 可以通过下面命令安装 cutecom/minicom 来使用 UART:
 - a) `sudo apt-get install cutecom`
 - b) `sudo apt-get install minicom`

联系我们

1. Github: <https://github.com/WeActTC>
2. 码云: <https://gitee.com/WeAct-TC>
3. 网站: <https://www.weact-tc.cn/>
4. 淘宝: <https://weactstudio.taobao.com/>



WeAct Studio
官方淘宝店